SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **MLA03**
COURSE CODE: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Data Interpretation **Duration: 1hr**

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent. **(5 Marks)**

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. **(5 Marks)**

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. **(5 Marks)**

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ε-greedy, softmax, and random exploration. The cumulative reward is highest with ε-greedy. Compare the strategies and defend why ε-greedy achieves better performance in this scenario.
(5 Marks)

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.
(5 Marks)

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.
(5 Marks)

Code of Conduct:

I T.chaitra(192425217)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:T.Chaitra

RUBRICS FOR CALCULATION

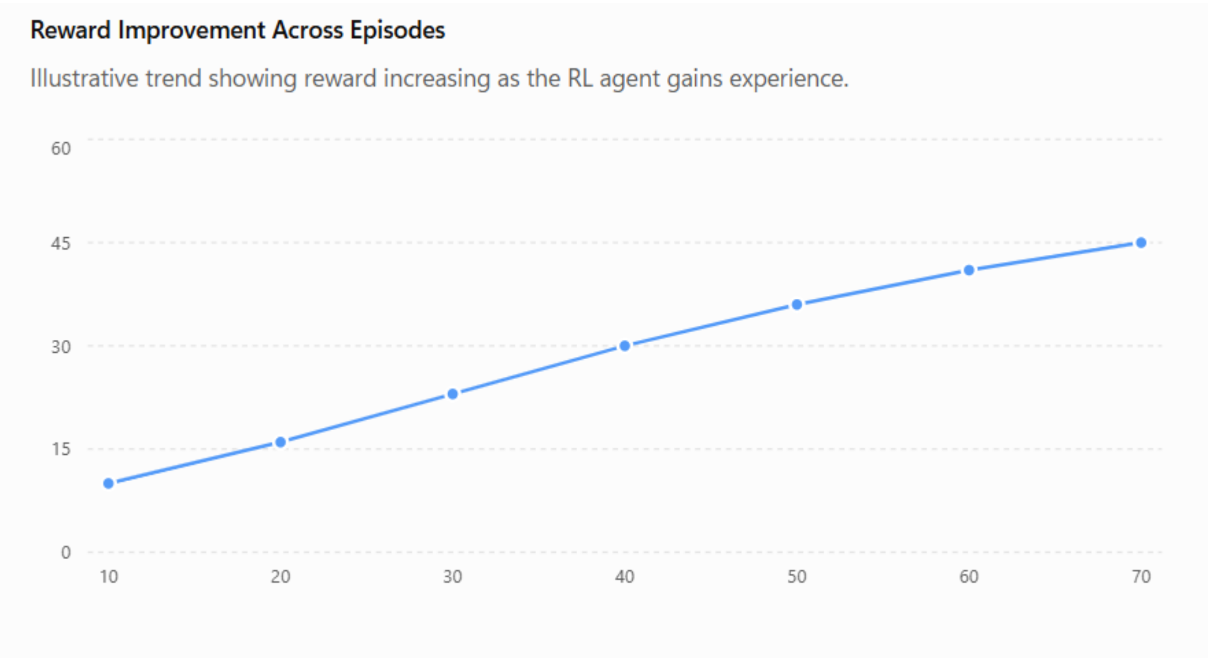
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

1. Reward increases from 10 to 45 over multiple episodes

Answer:
The reward values increase gradually from **10 to 45** as the number of training episodes increases. This indicates that the Reinforcement Learning agent is improving its performance through experience.
In the initial episodes, the agent has limited knowledge about the environment. Therefore, it may take inappropriate actions and receive lower rewards. As training continues, the agent learns from the rewards and penalties received after taking different actions. It gradually identifies which actions are more beneficial and starts following a better policy.
The increase from **10 to 45** represents a significant improvement in the agent's ability to make good decisions. A higher reward means that the agent is obtaining more desirable outcomes from its interactions with the environment.

This trend can be interpreted as:
More episodes → More experience → Better action selection → Better policy → Higher reward
If the reward continues increasing and eventually becomes stable around a high value, it suggests that the agent is approaching a good or optimal policy.

Conclusion:
The gradual increase in reward from **10 to 45** is a positive indication of learning. It shows that the agent is improving its policy through experience and becoming more effective at maximizing cumulative rewards. If the reward eventually stabilizes at a high level, it can indicate convergence toward an optimal policy.



2. Epsilon values 0.1, 0.3 and 0.5

Answer:

The epsilon value in an **epsilon-greedy strategy** controls the balance between **exploration and exploitation**.

- **Exploration:** The agent tries a new or random action to discover potentially better actions.
- **Exploitation:** The agent chooses the action that currently has the highest estimated value.

The epsilon value represents the probability of exploration.

Epsilon	Exploration	Exploitation
0.1	10%	90%
0.3	30%	70%
0.5	50%	50%

When **ε = 0.1**, the agent explores only 10% of the time and exploits the learned best action approximately 90% of the time. Therefore, if the agent has already learned a reasonably good policy, it can obtain high cumulative rewards.
When **ε = 0.3**, the agent performs more exploration. This may help discover new actions, but it also causes the agent to select some less rewarding actions.
When **ε = 0.5**, half of the decisions involve exploration. Therefore, the agent frequently takes actions that may not be optimal, resulting in the lowest cumulative reward in this experiment.

The observed relationship is:
ε = 0.1 → Highest reward
ε = 0.3 → Medium reward
ε = 0.5 → Lowest reward

Conclusion:

The best-performing epsilon value in this experiment is **0.1** because it provides a good balance with greater exploitation of the learned policy. However, epsilon should not always be very small during the early stages of training because sufficient exploration is necessary to discover good actions.

3. Q-Learning vs SARSA

Answer:

Q-Learning and SARSA are both **Temporal-Difference reinforcement learning algorithms**, but they differ in the way they update their Q-values.

- **Q-Learning** is an **off-policy** algorithm.
- **SARSA** is an **on-policy** algorithm.
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The given observation shows that Q-Learning consistently produces slightly higher average rewards than SARSA at **100, 200 and 300 episodes**.

For example, the general trend can be represented as:
Q-Learning reward > SARSA reward
at all three evaluation points.

Q-Learning updates its value using the maximum possible Q-value for the next state. Therefore, it learns the value of the best possible action, independent of the action actually selected during exploration.
SARSA, on the other hand, considers the actual next action selected by the current policy. Therefore, its learning can be more conservative, especially when exploration is present.

Since Q-Learning obtains higher rewards consistently, it demonstrates better performance under the given experimental conditions.

However, it is important to note that **higher reward alone does not mathematically prove faster convergence**. To determine convergence speed rigorously, we would normally examine how quickly the reward or Q-values become stable.

Based on the observed data, if Q-Learning reaches a high and stable reward earlier than SARSA, then Q-Learning can be considered to have converged faster.

Conclusion:
Q-Learning performs slightly better than SARSA in the given experiment because it consistently achieves higher rewards. The observed trend suggests that Q-Learning reaches a better-performing policy more quickly, although actual convergence speed should ideally be confirmed by examining reward stability and Q-value convergence.

4. Discount factors 0.2, 0.5 and 0.9

Answer:

The **discount factor (γ)** determines how much importance the agent gives to future rewards compared with immediate rewards.
The value of γ lies between **0 and 1**.

- A low γ gives more importance to immediate rewards.
- A high γ gives more importance to future rewards.
-

The given values are:

Discount factor	Importance of future rewards
$\gamma = 0.2$	Very low
$\gamma = 0.5$	Moderate
$\gamma = 0.9$	High

When $\boldsymbol{\gamma} = \mathbf{0.2}$, the agent focuses mainly on immediate rewards. It may ignore actions that provide small immediate rewards but lead to much greater rewards later.

When $\boldsymbol{\gamma} = \mathbf{0.5}$, the agent considers both immediate and future rewards to a moderate extent.

When $\boldsymbol{\gamma} = \mathbf{0.9}$, future rewards receive much greater importance. Therefore, the agent is encouraged to make decisions that may not provide the highest immediate reward but produce better long-term outcomes.

Since the average reward increases as γ increases, the environment appears to benefit from **long-term planning**.

The trend is:

$\gamma = 0.2$ → Lower reward
$\gamma = 0.5$ → Higher reward
$\gamma = 0.9$ → Highest reward

Conclusion:

The most suitable discount factor among the given values is **0.9**, because it produces the highest average reward. It allows the agent to consider long-term consequences and optimize cumulative future rewards. However, a very high discount factor can sometimes slow learning or be unsuitable for tasks where immediate rewards are more important.

5. Success rate increases from 45% to 92%

Answer:

The success rate increases from **45% at 50 episodes to 92% at 200 episodes**. This represents a substantial improvement in the agent's ability to complete the task successfully.

The improvement is:
92% − 45% = 47 percentage points

The success rate has therefore increased by **47 percentage points**.
At 50 episodes, the agent is still in the early learning stage. It has limited experience and may select inefficient or incorrect actions.
As training progresses to 200 episodes, the agent receives more feedback from the environment and improves its value estimates or policy. Consequently, it makes more appropriate decisions and achieves a higher percentage of successful outcomes.

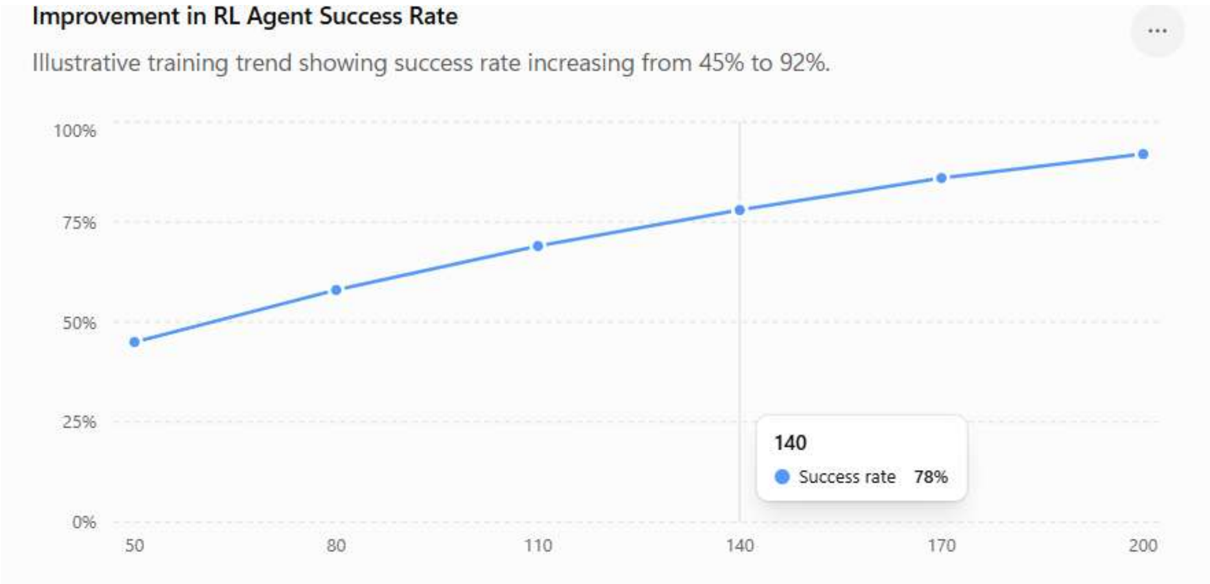
The trend can be represented as:

More training → More experience → Better policy → Higher success rate

A success rate of **92%** is very high and strongly suggests that the agent has learned an effective policy.
However, we should be careful before calling it definitively **optimal**. An optimal policy should ideally be verified by comparing the learned policy with the theoretically optimal policy or by observing that performance has converged and remains stable.

Conclusion:

The increase from **45% to 92%** demonstrates significant learning and policy improvement. The 92% success rate strongly indicates that the agent has learned a highly effective policy and may be approaching an optimal policy, particularly if the success rate remains stable in subsequent episodes.



6. Learning rates 0.01, 0.05 and 0.1 in CartPole

Answer:

The **learning rate (α)** determines how strongly newly obtained information changes the agent's existing knowledge.
A small learning rate means that the agent updates its knowledge slowly, whereas a large learning rate causes larger and faster updates.

The given learning rates are:

Learning rate	Effect
$\alpha = 0.01$	Slow learning
$\alpha = 0.05$	Balanced learning
$\alpha = 0.1$	Fast but potentially unstable

With **$\alpha = 0.01$**, updates are small. The agent may learn very slowly and require more episodes to reach a good policy.
With **$\alpha = 0.1$**, updates are much larger. This can allow rapid learning, but if the updates are too aggressive, the agent may overshoot good estimates and produce unstable learning.
With **$\alpha = 0.05$**, the agent achieves higher stability. This suggests that it provides a good balance between learning speed and stability.

Therefore:

0.01 → Stable but slow
0.05 → Stable and sufficiently fast
0.1 → Fast but potentially unstable

Conclusion:

The learning rate **0.05** provides the best trade-off between convergence speed and stability in the given CartPole experiment. It allows the agent to update its knowledge sufficiently quickly without causing excessive fluctuations in learning.

7. Epsilon-greedy vs Softmax vs Random exploration

Answer:
The three exploration strategies determine how an RL agent selects actions when it is uncertain about the environment.

1. Epsilon-greedy
Epsilon-greedy chooses:
- the best-known action most of the time;
- a random action with probability ε.

Therefore, it provides a simple and effective balance between exploration and exploitation.

2. Softmax exploration
Softmax assigns probabilities to actions based on their estimated Q-values. Actions with higher Q-values receive higher probabilities, while lower-valued actions still have some chance of being selected.
This provides more controlled exploration than completely random action selection.

3. Random exploration
Random exploration chooses actions without considering their learned values. Therefore, the agent may repeatedly select poor actions even after learning which actions are better.
Given that epsilon-greedy produces the highest cumulative reward, it is the best-performing strategy in this experiment.
The reason is that epsilon-greedy exploits the best-known action most of the time while still maintaining some exploration.

For example, with $\varepsilon = 0.1$:
90% exploitation + 10% exploration
This allows the agent to use its learned knowledge while occasionally searching for better alternatives.

Conclusion:
Epsilon-greedy achieves the highest cumulative reward because it provides a simple and effective balance between exploration and exploitation. Random exploration wastes many actions on potentially poor choices, while softmax may distribute exploration differently based on Q-values. Under the given experimental conditions, **epsilon-greedy is the most effective strategy**.

8. Maze steps reduce from 50 to 15

Answer:
The agent initially takes **50 steps** to reach the goal, but after 300 training episodes, it takes only **15 steps**.

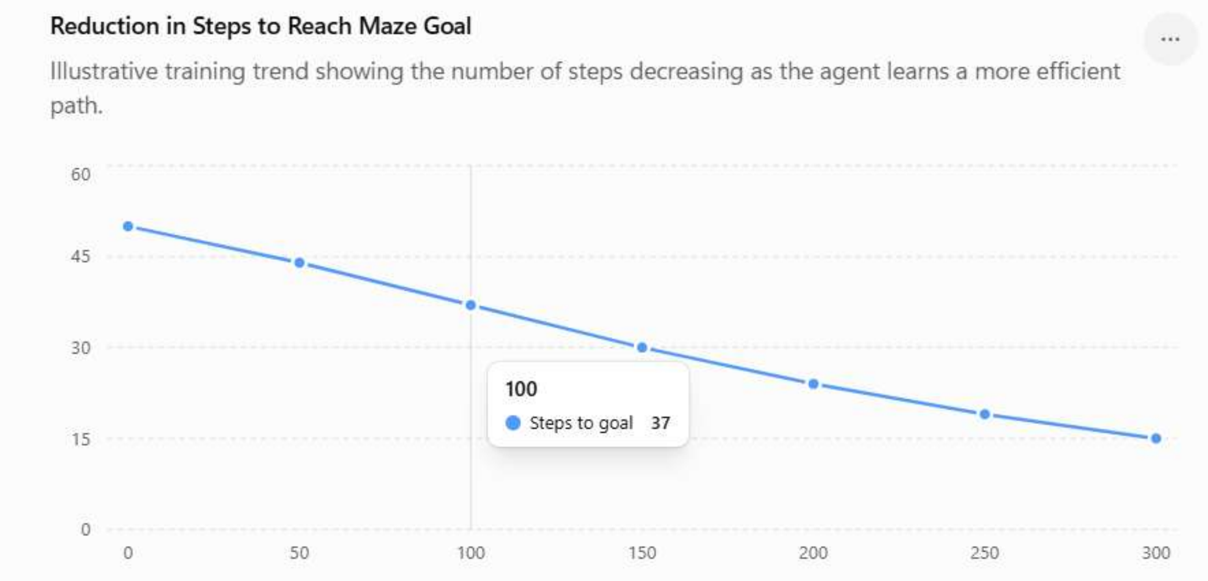
The reduction is:
50 − 15 = 35 steps
Therefore, the agent reduces the number of steps by **35 steps**.
The percentage reduction is:
Therefore, the number of steps decreases by approximately **70%**.
This is a strong indication that the agent has learned a more efficient route through the maze.
During early episodes, the agent does not know the best path. It may explore unnecessarily, move in wrong directions, or revisit the same states.
As training continues, the agent learns the value of different states and actions. It gradually identifies actions that lead closer to the goal.

The improvement can be represented as:

50 steps → Exploration and inefficient movement
↓
Training
↓
15 steps → Efficient path
If the number of steps remains close to 15 for additional episodes, this would provide stronger evidence of convergence.

Conclusion:

The reduction from **50 to 15 steps**, corresponding to a **70% reduction**, clearly demonstrates substantial policy improvement. It indicates that the agent has learned a much more efficient route. If the performance becomes stable around 15 steps and no shorter path exists, it can be considered evidence of convergence toward an optimal policy.



9. SARSA with γ = 0.4, 0.7 and 0.95

Answer:

The discount factor γ determines the importance given to future rewards in comparison with immediate rewards.
The three values are:

γ	Behaviour
0.4	Strong preference for immediate rewards
0.7	Moderate consideration of future rewards
0.95	Strong preference for future rewards

When **γ = 0.4**, the agent focuses mainly on immediate rewards. It may choose an action that gives an immediate benefit even if another action would produce a greater long-term reward.
At **γ = 0.7**, the agent considers both immediate and future rewards more effectively.
At **γ = 0.95**, future rewards receive very high importance. Therefore, the SARSA agent is encouraged to select actions that lead to better long-term outcomes.

The observed result is:
γ = 0.95 → Highest long-term reward
This suggests that the SARSA task requires significant long-term planning.

For example, in a maze, an action might temporarily move the agent away from the goal but eventually place it on a much shorter route. A high discount factor helps the agent recognize the long-term value of such actions.

Conclusion:

The discount factor **0.95** is most effective in this experiment because it strongly considers future rewards and produces the highest long-term reward. It is particularly suitable for environments where current decisions have important consequences for future states and rewards.

10. Fluctuating rewards initially, stable after 500 episodes

Answer:

The agent initially shows **fluctuating rewards**, but the rewards become more stable after approximately **500 episodes**.
This behaviour is common during reinforcement learning because the agent initially has limited knowledge of the environment and performs significant exploration.

During the early episodes:

- The agent does not know the optimal actions.
- It explores different states and actions.
- It sometimes receives high rewards.
- It sometimes receives low or negative rewards.
- Therefore, the reward curve fluctuates significantly.

As training progresses, the agent learns from its experiences and improves its policy. It gradually identifies actions that provide better outcomes.

After approximately **500 episodes**, the rewards stabilize. This suggests that the agent's policy is becoming consistent and that the agent is selecting effective actions more frequently.

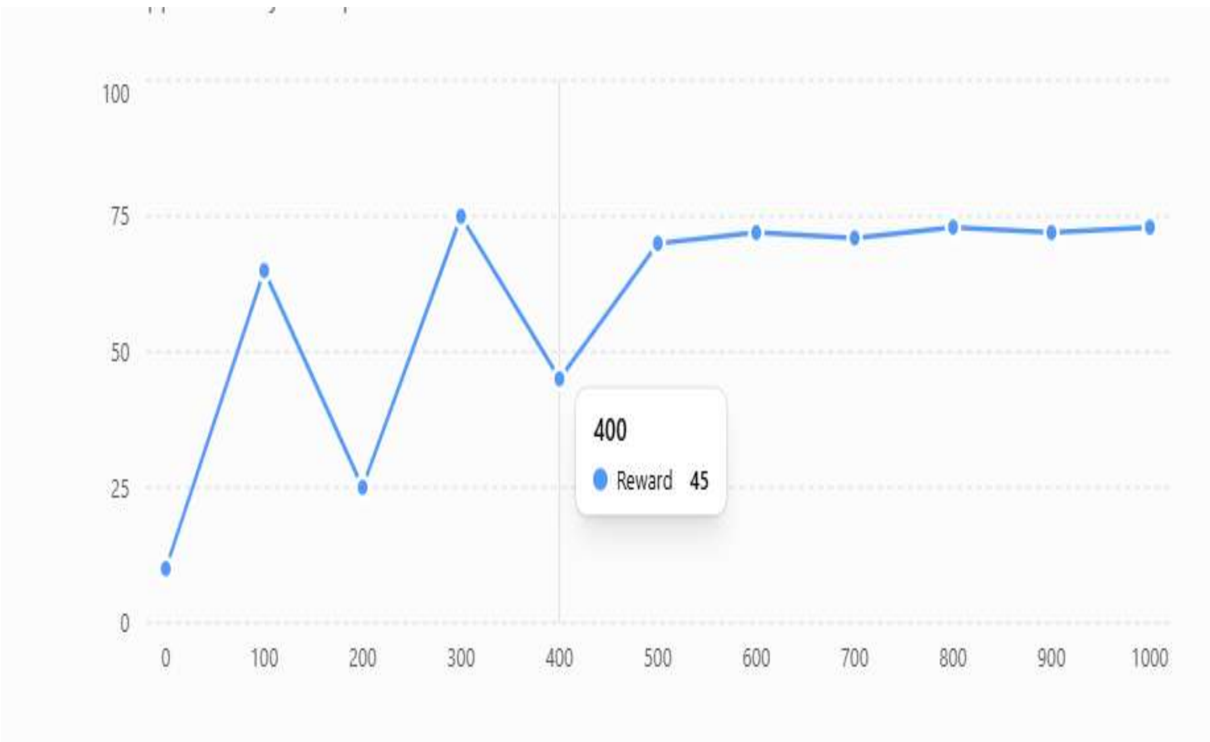
The trend can be represented as:

Early training
→ High exploration
→ Fluctuating rewards
→ Policy learning
Later training
→ Better knowledge
→ Reduced unnecessary exploration
→ Stable rewards
→ Improved policy

Stable rewards also indicate **robustness**. If the agent consistently obtains similar rewards rather than experiencing large fluctuations, its policy is becoming less sensitive to individual episodes.

However, stability alone does not guarantee that the policy is mathematically optimal. The stable reward must also be sufficiently high and should ideally be compared with an optimal or benchmark policy.

Conclusion:
The fluctuating rewards during the early episodes indicate active exploration and learning. The stabilization after **500 episodes** indicates that the agent has learned a more consistent policy and is approaching convergence. If the stable rewards are high and remain consistent over additional episodes, this provides strong evidence of policy improvement, convergence, and robustness.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